ISHANT BISHNOI

Machine Learning & Data Science Enthusiast

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PROFILE SUMMARY

MCA student at PES University (CGPA 8.57) with deep expertise in Machine Learning, Data Engineering, and Algorithmic Systems. Proven ability to transform raw, unstructured datasets into production-grade ML pipelines and actionable intelligence. C.N. Rao Merit Scholar with consistent academic excellence and a strong portfolio of real-world AI and data science projects.

INTERNSHIP EXPERIENCE

MetaWurks — **Apr 2025 – Present**

Software / ML Intern

- Currently contributing to ongoing projects and initiatives at MetaWurks.

cHeal — **Jan 2025 – Mar 2025**

Intern | 3 months

- Completed a 3-month internship, gaining hands-on industry experience in a professional environment.

EDUCATION

MCA – PES University, Bengaluru	2024 – 2026 (Ongoing)	**CGPA: 8.57 | C.N. Rao Scholar – Sem 1 & 3**
BCA – NIMS University, Jaipur	2021 – 2024	**CGPA: 8.51**
Class XII – RBSE	2021	**96.2%**
Class X – RBSE	2019	**84.83%**

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript

ML / AI: Scikit-learn, XGBoost, TensorFlow, Random Forest, SVM

Data Engineering: PySpark, Hadoop, Apache Flink, Pig Latin

Visualization: Tableau, Power BI, Matplotlib

Web Technologies: HTML5, CSS3, React

Libraries / Tools: Pandas, NumPy, pdfplumber, Git, SQLite

ACADEMIC PROJECTS

DAMA – Data-Driven Algorithmic Market Analysis

AI / ML | Python, XGBoost, Pandas, NumPy

- Architected a fully modular end-to-end pipeline for algorithmic analysis of **NSE-listed stocks** — from raw OHLCV ingestion to validated trade signal generation.
- Built the DAMA-Core eligibility engine using **EMA (10/20/50), Darvas Box, and ATR** to suppress noise and isolate only high-probability market setups.
- Integrated a tuned **XGBoost classifier** producing Buy / Sell / Hold signals with cross-validated backtesting on multi-year historical price data.
- Performed sector-wise stock classification and systematic feature engineering across large multi-year OHLCV datasets.

District-Level Disease Outbreak Prediction System

ML / Data Engineering | Python, XGBoost, Random Forest, pdfplumber

- Built an AI early-warning system spanning **700+ Indian districts**, processing **15+ years** (2009–2025) of IDSP government outbreak data.
- Automated extraction from unstructured government PDFs using **pdfplumber**, creating clean ML-ready structured datasets from scratch.
- Enriched feature sets with environmental signals — weather data, population density, and seasonal patterns — to boost predictive accuracy.
- Trained XGBoost and Random Forest ensembles to generate probabilistic district-level risk scores for early epidemic detection.

Expense Manager – Full Stack Application

Role: AI/ML & Analytics Module Lead | Python, ML, Dashboard

- Led the AI/ML analytics module of a group full-stack project for personal and group financial tracking across multiple users.
- Designed normalised expense datasets and built interactive dashboards visualising spending behaviour and category-level trends.
- Applied time-series ML forecasting to predict future expenses, improving budget planning accuracy by approximately **18%**.
- Structured the data pipeline for scalability, enabling seamless addition of new expense categories and user groups.

Interactive Chess Tournament Management System

Desktop Application | Python, Qt QML, SQLite, PyInstaller

- Built a full-featured **desktop application** managing tournaments end-to-end — player registration, automated pairings, round management, and real-time standings.
- Implemented **Swiss System** and **Round Robin** (Berger table) pairing algorithms with duplicate-opponent prevention and bye handling.
- Designed a **modular signal–slot architecture** (Qt QML frontend + Python backend + SQLite) for clean separation of concerns and event-driven UI updates.
- Packaged as a **standalone Windows executable** via PyInstaller; included CSV import/export, undo support, and timestamped database backup and restore.

Clothing Size Predictor

Machine Learning | Python, Scikit-learn, SVM, Pandas

- Built an **SVM classifier** to predict clothing size (XS, S, M, L) from body measurements — height, weight, chest, waist, and hip circumference.
- Applied feature normalisation (StandardScaler), handled class imbalance, and performed systematic **hyperparameter tuning** using GridSearchCV.
- Evaluated performance using **confusion matrix, per-class precision, recall, and F1-score**; identified misclassification patterns to guide further feature selection.
- Benchmarked against KNN and Decision Tree classifiers, demonstrating SVM's superior generalisation on the given measurement dataset.

CERTIFICATIONS & ACHIEVEMENTS

Certified Data Science – TutorialsPoint	In Progress
C.N. Rao Merit Scholarship – PES University	Awarded Sem 1 & Sem 3 for ranking among top academic performers
Hindustan Scout & Guide – State Award	State-level recognition for leadership and community service
Kabaddi – State Level	Represented school at state inter-district championship
Chess & Cricket – Academic Level	Regular participant in academic-level competitive events